

From Survey to Solution: Lightweight and Interpretable Multimodal Deepfake Detection Using CNNs and Grad-CAM

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Abstract—Deepfakes made with advanced deep learning methods are a big threat to the safety, privacy, and integrity of digital media. It's getting harder to tell when audio and video content has been changed because they look more real. Recent research has proposed various deepfake detection methods, ranging from unimodal approaches to multimodal frameworks that jointly analyse audio and visual information. However, many existing solutions are computationally expensive or lack interpretability, limiting their suitability for real-time and forensic applications. The state-of-the-art deepfake detection methods are thoroughly reviewed in this paper, with particular attention paid to model architectures, datasets, evaluation metrics, and significant limitations. Lightweight models, multimodal fusion techniques, and explainable artificial intelligence (XAI) methods are given special attention. A lightweight multimodal deepfake detection framework is proposed based on the identified research gaps. It uses Grad-CAM for explainable decision making, late fusion for multimodal integration, and convolutional neural networks for audio and video feature extraction. In order to make it appropriate for real-time deployment, the suggested framework seeks to strike an efficient balance between detection performance, computational efficiency, and interpretability. In addition, a structured experimental validation framework and computational efficiency analysis strategy are presented to facilitate practical implementation and benchmarking against existing multimodal techniques.

Keywords—Multimodal, CNN, GRAD-CAM, Deepfake, XAI, Late Fusion.

I. INTRODUCTION

Deepfakes, which are artificially altered audio, video, or image content intended to look real, have become increasingly common as a result of the quick development of deep learning and generative models. Although there are valid uses for these technologies in virtual communication and entertainment, their abuse poses major risks to social stability, digital trust, privacy, and security. Reliable detection mechanisms are a critical research priority because deepfakes are increasingly being used for political manipulation, identity fraud, disinformation campaigns, and the creation of non-consensual content.

Visual artifacts in manipulated videos, such as strange facial expressions, irregular eye blinking, and inconsistent blending, were the main focus of early deepfake detection techniques. When dealing with high-quality or compressed content, these video-based techniques frequently failed, despite achieving respectable performance in controlled

environments. Simultaneously, audio-based deepfake detection methods were developed to detect synthetic speech produced by text-to-speech and voice cloning systems. Single-modal methods are still susceptible, though, since attackers can control one modality while maintaining the relative authenticity of the other.

Recent research has moved toward multimodal deepfake detection, which simultaneously examines audio and video data, in order to get around these restrictions. By taking advantage of cross-modal inconsistencies, such as mismatched lip movements and speech patterns, multimodal systems provide increased robustness. Despite their efficacy, a number of multimodal approaches currently in use rely on computationally costly architectures, such as transformers and large-scale attention models, which limit their use in real-time and resource-constrained settings.

Although transformer-based multimodal systems have a high detection rate, their enormous parameter size and attention-based computations considerably increase memory and inference overhead. This limits their use in edge devices and real-time forensics systems. As a result, a computationally efficient but interpretable multimodal solution is necessary.

The inability of existing deepfake detection systems to be interpreted is another significant drawback. The majority of deep learning models operate as "black boxes," producing classification results without providing an explanation for their choices. In high-stakes applications where explainability and trust are crucial, like digital forensics, law enforcement, and content moderation, this lack of transparency is especially problematic. Explainable artificial intelligence (XAI) methods that can identify the areas or characteristics affecting a model's prediction are therefore in greater demand.

To address these issues, this paper provides a thorough analysis of deepfake detection techniques, including explainable methods, lightweight architectures, and single-modal and multimodal approaches. A lightweight and comprehensible multimodal deepfake detection framework is also suggested based on the survey's findings. The framework uses a fully connected layer for classification, late fusion to combine multimodal representations, and Convolutional Neural Networks (CNNs) for effective audio and video feature extraction. Gradient-weighted Class Activation Mapping (Grad-CAM) is used to provide visual explanations for both audio and video predictions in order to improve transparency.

This paper makes three main contributions:

1. A structured review of existing deepfake detection methods with emphasis on multimodality, efficiency, and explainability
2. identification of key research gaps in lightweight and interpretable deepfake detection and
3. the proposal of a practical multimodal framework suitable for real-time and trustworthy deployment.

II. LITERATURE REVIEW

A. Video-Based Deepfake Detection

Video-based deepfake detection methods analyse spatial and temporal inconsistencies introduced during face manipulation. Early studies relied on handcrafted features such as eye blinking frequency, facial landmarks, and head pose estimation to detect unnatural patterns. These approaches provided good interpretability but lacked robustness against high-quality deepfakes.

With the adoption of deep learning, CNN-based architectures became widely used for extracting discriminative visual features from facial regions. Shallow CNNs such as MesoNet demonstrated that lightweight architectures could achieve reasonable accuracy with low computational overhead. However, deeper CNNs and temporal models, including 3D CNNs and CNN-RNN hybrids, were later introduced to capture motion-based inconsistencies. Although these methods improved detection accuracy, they significantly increased model complexity and inference time.

B. Audio-Based Deepfake Detection

Audio deepfake detection focuses on identifying synthetic speech generated by voice conversion and text-to-speech systems. Traditional signal-processing approaches analysed spectral features and prosodic cues, but these methods were insufficient for detecting modern, high-quality synthetic audio.

Recent works employ CNNs trained on Mel-spectrograms or MFCC representations to capture time-frequency artifacts unique to synthetic speech. CNN-based audio detectors have shown strong performance and relatively low computational cost. Nevertheless, audio-only detection remains vulnerable when the visual modality is manipulated independently or when high-fidelity voice cloning is used.

C. Multimodal Deepfake Detection

Multimodal deepfake detection aims to improve robustness by jointly analysing audio and video modalities. These approaches exploit cross-modal inconsistencies such as lip-speech mismatch and audiovisual synchronization errors. Fusion strategies are typically categorized as early fusion, late fusion, or hybrid fusion.

While transformer-based multimodal frameworks with cross-modal attention have demonstrated high detection accuracy, they require substantial computational resources and large training datasets. In contrast, late fusion methods that combine modality-specific features at the decision level offer a more efficient alternative, though they remain underexplored in lightweight and explainable settings.

D. Lightweight and Explainable Approaches

The need for real-time deployment has motivated research into lightweight deepfake detection models. Compact CNN

architectures and mobile-optimized networks reduce parameter count and inference latency while maintaining acceptable performance. However, many lightweight models compromise interpretability and provide limited insights into decision-making.

Explainable AI techniques, particularly Grad-CAM, have been applied to CNN-based deepfake detectors to visualize regions influencing classification. In video-based detection, Grad-CAM highlights manipulated facial regions, while in audio-based detection, it identifies discriminative time-frequency components in spectrograms. Despite their effectiveness, explainability is rarely integrated into multimodal deepfake detection frameworks.

E. Recent Advances (2024-2026)

Recent research has concentrated on transformer-based multimodal architectures and cross-modal attention techniques that improve semantic consistency between audio and visual streams. While these models improve detection robustness and cross-dataset generalization, they also add significant computational complexity. In parallel, lightweight CNN-based models for real-time and edge deployment situations have been developed, with an emphasis on low parameter count and efficient inference. These findings highlight the trade-off between detection performance and computing efficiency, emphasizing the importance of organized lightweight multimodal frameworks.

F. Comparative Summary of Literature

TABLE I presents a comparative summary of representative deepfake detection methods, categorized by modality, architecture, strengths, and limitations. The table shows that existing techniques prioritize either high detection accuracy through sophisticated multimodal architectures or computational efficiency via lightweight CNN models. However, integrated multimodal explainability within computationally efficient frameworks is still underexplored.

TABLE I: SUMMARY OF DEEPAKE DETECTION METHODS

Author / Year	Modality	Method	Key Contribution	Advantages	Limitations
Afchar et al. (2018)	Video	CNN (MesoNet)	Detection of visual manipulation artifacts	Lightweight and fast	Limited generalization
Li et al. (2018)	Video	Eye blinking analysis	Detection of temporal inconsistencies	Interpretable features	Ineffective on newer deepfakes
Guera & Delp (2019)	Video	CNN-RNN	Spatio-temporal deepfake modelling	Improved robustness	High computational cost
Yang et al. (2019)	Video	Head pose analysis	Geometric inconsistency detection	Explainable cues	Sensitive to pose variations
Wang et al. (2020)	Audio	CNN on spectrograms	Learning audio manipulation artifacts	Efficient feature learning	Audio-only limitation

ASVspoof (2021)	Audio	CNN-based models	Benchmark for synthetic speech detection	Standardized evaluation	No visual information
Mittal et al. (2022)	Multimodal	CNN-based fusion	Audio-video correlation modelling	Improved accuracy	Heavy architecture
Wu et al. (2023)	Multimodal	Transformer fusion	Cross-modal attention learning	State-of-the-art performance	Computationally expensive
XAI-based (2024)	Audio / Video	CNN + Grad-CAM	Explainable deepfake detection	Improved transparency	Limited multimodal focus
Du et al. (2025)	Audio-Video	Cross-modal alignment & distillation	Harmonized multimodal representation	Better fusion strategy	Complex training process
Wu et al. (2025) (HOLA)	Audio-Video	Hierarchical contextual aggregation	Enhanced spatio-temporal modelling	Strong performance	Large-scale pre-training required
Sun et al. (2025) (EDVD-LLM)	Video + XAI	Multimodal reasoning model	Explainable deepfake reasoning	High interpretability	Resource intensive
Kundu et al. (2025) (TruthLens)	Video	Hybrid local-global modelling	Explainable hybrid detection	Improved transparency	Increased architecture complexity
Sun et al. (2026)	Audio-Video	Semantic consistency fusion	Cross-modal semantic alignment	Improved robustness	Increased computational overhead
Proposed Framework	Audio-Video	CNN + Late Fusion + Grad-CAM	Lightweight and interpretable multimodal detection	Efficient and transparent	Requires synchronized inputs

G. Research Gap

TABLE I summarizes contemporary multimodal deepfake detection models that improve resilience via semantic consistency modeling and transformer-based fusion procedures. However, these structures add greatly to computational complexity and training requirements. Lightweight CNN-based techniques minimize model size, but they frequently lack structured multimodal fusion and embedded interpretability. Furthermore, explainability methodologies are usually used as post-hoc analysis rather than being incorporated into architectural design. These findings highlight the need for a lightweight, computationally efficient multimodal deepfake detection framework with structured explainability suitable for real-world use.

III. PROPOSED METHODOLOGY AND ARCHITECTURE

This section describes the proposed lightweight and interpretable deepfake detection framework, designed to

operate under flexible input conditions. The framework supports **audio-only, video-only, and combined audio-video inputs**, enabling robust detection across diverse real-world scenarios. The proposed methodology emphasizes efficient feature extraction, multimodal learning, and explainability, making it suitable for real-time and resource-constrained environments.

A. System Overview

The proposed framework employs modality-specific feature extraction pipelines that can operate independently or jointly depending on the availability of input modalities. When both audio and video streams are present, a late fusion strategy is applied to combine high-level representations. In unimodal scenarios, the framework performs classification using the available modality alone.

Overall Processing Pipeline:

1. The input sample may consist of audio-only, video-only, or combined audio-video data.
2. If video data is available, frames are sampled and facial regions are detected and aligned using MTCNN.
3. If audio data is available, the audio signal is transformed into time-frequency representations.
4. CNN-based models extract features independently from the available modalities.
5. When both modalities are present, a late fusion mechanism combines multimodal features.
6. A fully connected layer performs real/fake classification.
7. Grad-CAM generates explainable visualizations for the model's predictions.

The overall architecture of the proposed framework is illustrated in FIG. 1, showing parallel audio and video pipelines with multimodal fusion, followed by explainable classification.

B. Video Feature Extraction Module

For video-based inputs, frames are uniformly sampled and processed through a facial preprocessing stage. Facial regions are detected and aligned using **Multi-task Cascaded Convolutional Networks (MTCNN)** to reduce background interference and enhance feature quality. The aligned facial frames are then passed to a lightweight CNN for spatial feature extraction.

C. Audio Feature Extraction Module

For audio-based inputs, the raw audio signal is converted into Mel-spectrogram representations and processed using a CNN-based audio feature extractor. This module captures spectral anomalies associated with synthetic or manipulated speech.

D. Multimodal Late Fusion Strategy

When both audio and video modalities are available, the proposed framework applies a late fusion strategy by concatenating the extracted audio and video feature vectors. This unified representation enables complementary learning across modalities and improves robustness against modality-specific manipulation.

In unimodal scenarios, the fusion stage is bypassed, and classification is performed using the available modality alone.

E. Classification Module

The feature representation - either unimodal or multimodal - is passed through a lightweight fully connected layer followed by a SoftMax activation function to perform binary classification between real and deepfake samples.

F. Explainability Using Grad-CAM

Grad-CAM is applied to the convolutional layers of the active modality-specific CNNs. For video inputs, Grad-CAM highlights facial regions influencing the decision, while for audio inputs, it visualizes discriminative time-frequency regions in the spectrogram. In multimodal cases, explanations are generated independently for both modalities.

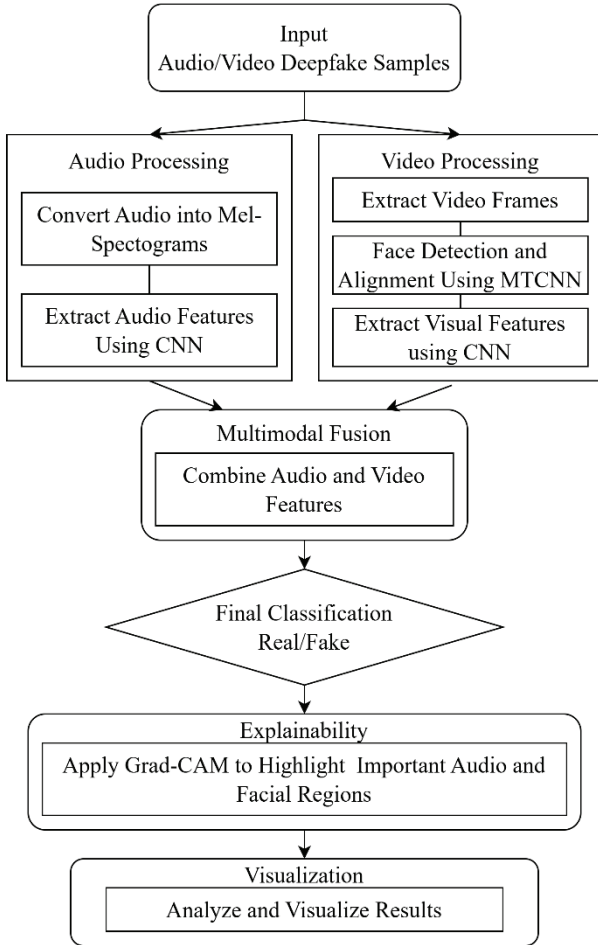


Fig. 1. Proposed Framework

G. Key Characteristics of the Proposed Architecture

The proposed framework is characterized by:

- Flexible input support (audio-only, video-only, or audio-video)
- Lightweight CNN-based design
- Robust facial preprocessing using MTCNN
- Multimodal fusion when applicable
- Explainability through Grad-CAM
- Scalability for real-time and edge deployments

IV. EXPERIMENTAL EVALUATION PLAN

Implementation and experimental evaluation of the suggested framework are presently underway to confirm its efficacy. To guarantee systematic benchmarking, the evaluation process listed below has been established.

A. Datasets

The system will be tested using publicly accessible benchmark datasets using a variety of processing methods and compression scenarios, including FaceForensics++, Celeb-DF, and DFDC.

B. Metrics for Evaluation

Standard measures such as Accuracy, Precision, Recall, F1-score, and AUC-ROC will be used to evaluate performance. Class-wise detection performance will also be assessed using confusion matrix analysis.

C. Analysis of Computational Efficiency

The following metrics will be assessed in order to support the lightweight design:

- Total number of parameters
- Size of the model (MB)
- Time spent inference per frame
- GPU memory use

D. Analysis via Comparison

The suggested framework will be contrasted with:

- Unimodal CNN-based models
- Transformer-driven multimodal systems
- Current models for explainable detection

E. Analysis of Robustness

Robustness will be assessed in light of:

- Compression of videos
- Noise that is additive
- Testing across datasets

V. CONCLUSION AND FUTURE WORK

This paper provided a structured examination of current deepfake detection approaches, emphasizing the growing necessity for multimodal analysis and integrated interpretability. According to the review, unimodal systems frequently encounter robustness and generalization issues when confronted with advanced manipulation techniques, whereas recent multimodal and transformer-based models, while effective, add significant computational complexity.

To overcome these issues, a lightweight and understandable multimodal deepfake detection system was presented. The architecture includes separate CNN-based audio and video processing pipelines, MTCNN-based face area extraction for accurate visual feature representation, and a late fusion technique for flexible modality merging. Grad-CAM allows for structured visual explanations of model predictions, which improves transparency and user trust.

At the time of submission, the suggested framework was still being implemented and experimentally validated. A complete evaluation strategy has been developed to evaluate

detection performance, computing efficiency, and robustness across multiple benchmark datasets. Future research will concentrate on empirical benchmarking, real-time optimization, and the development of enhanced fusion and explainability algorithms to improve scalability and deployment feasibility in practical deepfake detection systems.

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